

MAHDIYA RAHMAN SUKANYA

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[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

RESEARCH INTERESTS

Reliable AI Systems, Formal Methods, Human–AI Interaction, Human–Computer Interaction, Model Optimization

EDUCATION

Bachelor of Science in Software Engineering

Jan 2022–Oct 2026

Institute of Information Technology, University of Dhaka

CGPA: 3.50/4.00

Dhaka, Bangladesh

(Through 7th semester)

Higher Secondary Certificate (HSC)

2021

Murari Chand College, Sylhet, Bangladesh

GPA: 5.00/5.00

- Awarded Government Scholarship for Outstanding Academic Excellence

Secondary School Certificate (SSC)

2019

Blue Bird School and College, Sylhet, Bangladesh

GPA: 5.00/5.00

- Awarded Government Scholarship for Outstanding Academic Excellence

Junior School Certificate (JSC)

2016

Blue Bird School and College, Sylhet, Bangladesh

GPA: 5.00/5.00

- Awarded Government Scholarship for Outstanding Academic Excellence

RESEARCH ACTIVITIES

Conference and Workshop Proceedings

- Wasi, A. T., Khan, M. A., Rahman, A., Faisal, W., Saha, S., Bhuiyan, S., **Rahman Sukanya, M.**, Priti, R. N., Hoque, M. I., et al. (2026). InvariantBench: Can Large Language Models Exhibit Inherent Reasoning Across Equivalent Transformations? In **ICML 2026 Workshop AI4Math**. [\[Paper\]](#)
- Bhuiyan, S., **Rahman Sukanya, M.**, Hossen Roni, M. K., & Bin Mohsin, M. (2026). Neuro-Sync Adaptive Compression (NSAC): A Multi-Stage Pipeline for LLM Inference at Low Resource Environment. In **Proceedings of IRAI 2026**.
- Wasi, A. T., & **Rahman Sukanya, M.** (2026). Human–AI Interaction in Low-Tech, Low-Resource Healthcare Systems. In **ACM IH 2026**. [\[Paper\]](#)

Research Ongoing

- **Rahman Sukanya, M.**, Bhuiyan, S. & Shoyaib, M. AI-Driven Input Field Analysis and Dynamic UI Traversal for Automated Mobile Application Testing via the TestCube Framework (manuscript in preparation).
- **Rahman Sukanya, M.** & Khaled, S. M. VentAssist: An Offline Reinforcement Learning Ventilator Decision Support System (manuscript in preparation).

AWARDS & ACHIEVEMENTS

- **Finalist**, IEEE IES Generative AI Challenge 2026
- **Merit Award & Top 8 Globally**, International Data Science Olympiad (IDSOL), AI Track, Hong Kong 2025
- **Finalist (Top 5) & Honorable Mention**, National Blockchain Olympiad (AI Track) 2024
- **International AI Olympiad Participant**, Amsterdam, Netherlands 2024

RESEARCH EXPERIENCE

Summer Research Intern

2026

UIUC++ Summer Research Program in Software Engineering

- Benchmarked production failures across real-world software systems to evaluate autonomous software engineering agent behavior, supervised by Prof. [Tianyin Xu](#) (UIUC).
- Worked on the TLAPS Proof Benchmark project to evaluate AI models' ability to generate formally correct,

mechanically verified TLA+ proofs for testing formal reasoning capabilities.

Research Student

Computational Intelligence and Operations Lab (CIOL)

- Investigated human–AI interaction, AI-assisted HCI/HMI systems, and ergonomics, bias, and ethical challenges in interaction design, publishing a short paper at ACM on this work.

Undergraduate Researcher, TestCube Research Project

Institute of Information Technology, University of Dhaka

- Conducted research on AI-driven input field analysis and dynamic UI traversal for automated mobile app testing using TestCube, supervised by Dr. Mohammad Shoyaib, and actively working on publishing a paper on this work.

PROFESSIONAL EXPERIENCE

AI R&D Associate

Oct 2025–Present

Sigmoix AI

- Led AI-driven innovations spanning custom RAG pipelines, voice-based intelligent agents, and data-informed automation systems designed to transform raw information into actionable insights.

Software Engineering Intern

Nov 2025–Mar 2026

Streams Tech

- Executed comprehensive manual and automated testing of production-level software, identifying and documenting critical defects across system components.
- Engineered automated test suites that strengthened regression coverage and improved release confidence across the development pipeline.

PROFESSIONAL SERVICE

Judge and Question Setter, Bangladesh AI Olympiad

2026

- Designed questions for both regional and national rounds, and served as a judge evaluating participant solutions during the national round.

SKILLS

- **AI/ML:** PyTorch, Hugging Face, RAG, RL, Model Compression
- **Languages:** Python, C/C++, Java, JavaScript/TypeScript, SQL
- **Tools:** FastAPI, React, Next.js, Docker, AWS, Git/GitHub

LEADERSHIP EXPERIENCE

IIT Girls Community (IITGC) — General Secretary

2025–Present

A community for women in technology at IIT, University of Dhaka

- Spearheading the planning and execution of a national-level girls-only tech event encompassing a hackathon, competitive programming contest, and empowerment initiatives.

IIT Software Engineers' Community (IITSEC) — General Member

2024–2025

A student community for software engineers at IIT, University of Dhaka

- Contributed to organizing Flutter Frenzy (Google-sponsored conference), coordinating logistics and outreach.
- Supported the execution of ITVerse, a national-level IT competition spanning hackathons, project showcases, and competitive programming.